ALEK FRÖHLICH

Ph.D. Student in Machine Learning

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EDUCATION

ELLIS Ph.D. in Computer Science — Nov 2024 – Present
Istituto Italiano di Tecnologia & École Polytechnique — Genoa, Italy

- Leveraging low-rank spectral structure for scalable and statistically rigorous causal inference with high-dimensional data.
- Advisors: Massimiliano Pontil, Karim Lounici, Vladimir Kostić.

M.Sc. in Applied Mathematics — Mar 2022 – Jul 2024
Federal University of Santa Catarina — Florianópolis, Brazil

- GPA: 9.76/10. Courses: Functional analysis, machine learning theory, numerical optimization, differential equations, topology, advanced analysis, linear algebra and matrix theory.
- Dissertation: Elements of Learning theory and their Application in the Prediction of Malignancy of Breast Lesions.

B.Sc. in Computer Science — Mar 2018 – Mar 2022
Federal University of Santa Catarina — Florianópolis, Brazil

- GPA: 9.23/10. (2nd in cohort). Some courses: Measure theory, machine learning, probability, statistics, parallel programming, algorithms, software engineering, operating systems.

Exchange Student — Aug 2016 - Dec 2016
University High School — Irvine, CA

- AGPA: 4.2. AP Course: Computer science.

EXPERIENCE

Research Intern — Jul 2024 – Sep 2024
Federal University of São Carlos — São Carlos, Brazil

- Developed an interpretable breast cancer risk assessment tool with local coverage uncertainty quantification guarantees via conformal prediction. Resulted in [4]. Mentor: Rafael Izbicki.

Research Intern — Jul 2023 – Jul 2024
University of São Paulo Medical School — Ribeirão Preto, Brazil

- Developed and validated machine learning models for predicting malignancy of suspicious breast lesions identified during ultrasound examinations. Resulted in [5].
- Developed a computational pathology pipeline for segmenting and classifying tumor cells in whole slide images. Resulted in [6]. Mentor: Daniel Tiezzi.

Visiting Student — Jan 2023 – Nov 2023
Institute for Pure and Applied Mathematics (IMPA) — Rio de Janeiro, Brazil

- Courses: Functional analysis and machine learning theory.
- Seminars: Optimization (mentored by Benar Svaiter) and physics-informed neural networks.

Software Engineer Intern Jan 2019 – Dec 2019
LabSEC & Kryptus Florianópolis, Brazil

PUBLICATIONS & PRE-PRINTS

Symbols: † denotes equal contribution, * denotes highlight.

1. * **A Fröhlich**, V Kostić, K Lounici, D Perazzo, M Pontil. *Toward Scalable and Valid Conditional Independence Testing with Spectral Representations*. 2025. [arXiv]
2. * D Meunier,† J Wornbard,† V Kostić,† A Moulin, **A Fröhlich**, K Lounici, M Pontil, A Gretton. *Outcome-Aware Spectral Feature Learning for Instrumental Variable Regression*. 2025. [arXiv]
3. D Ordoñez-Apraez, V Kostić, **A Fröhlich**, V Brandt, K Lounici, M Pontil. *Equivariant Representation Learning for Symmetry-Aware Inference with Guarantees*. 2025. [arXiv]
4. **A Fröhlich**,† T Ramos,† G Santos, I Buzatto, R Izbicki, D Tiezzi. *PersonalizedUS: Interpretable Breast Cancer Risk Assessment with Local Coverage Uncertainty Quantification*. AAAI 2025. [DOI]
5. I Buzatto, S Recife, L Miguel, R Bonini, N Onari, A Faim, L Silvestre, D Carlotti, **A Fröhlich**, D Tiezzi. *Machine Learning can Reliably Predict Malignancy of Breast Lesions based on Clinical and Ultrasonographic Features*. Breast Cancer Research and Treatment 2024. [DOI]
6. D Tiezzi, **A Fröhlich**, F Chahud, S Pagnota. *197P Computational pathology pipeline enables quantification of intratumor heterogeneity and tumor-infiltrating lymphocyte score*. ESMO Immuno-Oncology and Technology 2023. [DOI]

TALKS

Spectral Representation Learning Oct 2025
University of Genoa Genoa, Italy

Spectral Representation Learning for High-dimensional Inference Aug 2025
Federal University of São Carlos São Carlos, Brazil

Breast Cancer Risk Assessment with Local Coverage UQ Jun 2025
New Trends in Statistical Learning V Porquerolles, France

Machine Learning Methods in Her2+ Breast Cancer Oct 2023
Institute for Pure and Applied Mathematics (IMPA) Rio de Janeiro, Brazil

AI and Her2+ Breast Cancer Aug 2023
University of São Paulo Medical School Ribeirão Preto, Brazil

SVM: Optimization Problem, Learning Guarantees, and Kernel Trick Jul 2023
34th Brazilian Mathematical Colloquium Rio de Janeiro, Brazil

SUPERVISIONS

Enzo N. Spotorno (B.Sc. co-advised with A. A. Fröhlich) Jul 2024 – Jul 2025
Federal University of Santa Catarina
Project: Improving Battery End-of-Life Prediction with Hybrid Bayesian PINNs

Vagner S. Santos (M.Sc. co-advised with R. Izbicki) Mar 2026 – Mar 2028
Federal University of São Carlos
Project: Spectral Boosting for Nonparametric Causal Inference

AWARDS & RECOGNITIONS

INdAM Membership (Real Analysis, Measure Theory & Probability) 2026
Istituto Nazionale di Alta Matematica (INdAM) Italy

- National network of mathematical excellence supporting research projects and mobility.

CAPES Master's Scholarship 2022 – 2024
Brazilian Federal Agency for Support and Evaluation of Graduate Education (CAPES) Brazil

- Fully funded my M.Sc. studies in Applied Mathematics (R$ 50,400).

FAPEU Scholarship 2019
Fundação de Amparo à Pesquisa e Extensão Universitária (FAPEU) Brazil

- Awarded for developing the access control system for UFSC's university restaurant.

FEESC Scholarship 2019
Fundação Stemmer para Pesquisa, Desenvolvimento e Inovação (FEESC) Brazil

- Awarded for work on the Cryptographic Service Provider developed at LabSEC.

ITI Scholarship 2019
Instituto Nacional de Tecnologia da Informação (ITI) Brazil

- Awarded for work on the Cryptographic Service Provider developed at LabSEC.

OUTREACH & LEADERSHIP

Reviewer for ICLR 2026 Workshop: Catch, Adapt, and Operate Feb 2026

EVENTS

Causality for Impact Workshop EurIPS 2025 Dec 2025
Oral Presentation [1] Copenhagen, Denmark

Mathematical Aspects of Data Science Summer School Sep 2025
École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland

Workshop on Making Sense of Data in Robotics CoRL 2025 Sep 2025
Presentation of [3] Seoul, Korea

Theoretical Foundations of Machine Learning Summer School Jun 2025
University of Genoa Genoa, Italy

New Trends in Statistical Learning V Jun 2025
Presented [4] Porquerolles, France

39th Annual AAAI Conference on Artificial Intelligence Feb 2025
Presented [4] Philadelphia, PA

Workshop on Optimization and Inverse Problems 2023
Federal University of Santa Catarina Florianópolis, Brazil

34th Brazilian Mathematical Colloquium 2023
Institute for Pure and Applied Mathematics (IMPA) Rio de Janeiro, Brazil

Machine Learning and Combinatorics Workshop 2020
Moscow Institute of Physics and Technology Moscow (online), Russia

Brazilian Symposium on Computing Systems Engineering IX 2019
Federal University of Rio Grande do Norte Natal, Brazil

The Developer's Conference 2019
Largest event related to software development in Brazil Florianópolis, Brazil

Chip in the Pampa 2018 2018
Federal University of Pelotas Pelotas, Brazil

Workshop on Quantum Computing I 2018
Federal University of Santa Catarina Florianópolis, Brazil

SKILLS

Languages	Portuguese (native), English (fluent), Italian (basic)
Programming	C/C++, Java, Python, R, Matlab, PyTorch
Miscellaneous	Git, Docker, LaTeX, SQL, Assembly (x86, ARM, RISC-V, MIPS)